

Tushar Tyagi

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EDUCATION

M.S. Computer Science — Georgia Institute of Technology	Aug 2024 – Dec 2026
GPA: 4.0 / 4.0 Specialization: ML Research: Llamas Lab TA: MGT 6655 Business Data Prep & Viz	
B.Tech Computer Science & Engineering — IIT Mandi	Aug 2017 – Jun 2021
CGPA: 8.39 / 10.00 Minor: Management Studies TA: IC152 Intro to Python & Data Science	

PATENT

[US20240404070A1](#): Precise book page boundary detection using deep ML model and image processing algorithms.

EXPERIENCE

Machine Learning Engineer 2 · Adobe, Noida, India	Jul 2021 – Jul 2025
Acrobat AI Assistant (Actions): Architected modular LLM-based actions pipeline; applied Chain-of-Thought prompting for 40% query-accuracy improvement and 66% performance gain; delivered real-time agentic troubleshooting solution. [Prompt Engineering, Agentic AI, Modular RAG] Document Classification Pipeline: Engineered real-time computer vision image classification model with 90% accuracy, boosting feature MAU by 50%; developed custom downsampling algorithm achieving 10× speedup over Lanczos in C++; curated, solved data imbalance and preprocessed large-scale labeled datasets using LabelMe. [Python, TensorFlow, PyTorch, OpenCV, TFLite, C++] Performance Optimization: Eliminated 3-second GPU loading delay via pre-loading pipeline; enhanced boundary detection by 25.6%; revitalized 3M+ dormant users; achieved 3× inference speedup with 4× model-size reduction via quantization & pruning. [TFLite, ONNX Runtime] Book Mode Feature: Spearheaded dual-page capture adopted by 5M monthly users; boosted segmentation accuracy by 20%, doubled throughput, improved dewarping accuracy from 70% to 80%. [Python, C++, TensorFlow, Precise Image Segmentation, U-Net] Form Detection Model: Built segmentation model achieving 70% beta accuracy; conducted downsampling algorithm experiments for pipeline optimization. [Python, C++, TensorFlow]	
Product Development Intern · Adobe, Noida, India	May 2020 – Jul 2020
- Built intelligent image-cleaning tools using Hough-Lines Transformation in OpenCV and deep learning (Keras). - Authored VBA scripts in Word to generate synthetic training data for segmentation models.	

PROJECTS

- VASTRA: Deterministic Agentic Video Reranking:** Architected a query-complexity-aware video retrieval system compiling NL queries into structured predicate programs; implemented uncertainty-gated visual verification with tiered candidate pruning to eliminate non-deterministic LLM loops. Achieved up to **6.4× faster latency** and **strictly deterministic rankings** ($\sigma = 0.000$), matching/exceeding ReAct baselines. [Python, asyncio, FAISS, VLM, Multimodal RAG, Agentic Systems]
- LLM Fine-Tuning (GRPO):** Implemented Group Relative Policy Optimization for LLM alignment; used group-mean rewards as control variates and KL-divergence penalties to stabilise policy updates. [Reinforcement Learning]
- VLM Context Engineering for Video Analytics:** Task-type classifiers dynamically select among five RAG-to-agentic retrieval strategies, achieving **10% accuracy gain** and **2× speedup** over static methods on Video-MME. [LangGraph, Python, Agentic AI Systems, Multimodal RAG, Pydantic]
- PROPEL:** Training-free framework using empirical bias profiling and inverse propensity weighting to mitigate position bias in LLM re-ranking; improves NDCG@1 by up to **26.7%** on real-world datasets. [LLM based Recommendation Systems, Python]
- Historical Manuscript OCR** [Open-Source AI Contribution]: Evaluated 8 Vision-Language Models (Qwen 2B–32B) for handwritten text recognition; achieved **0.124 CER** with Qwen2.5-VL-7B. Applied LoRA fine-tuning for **52% CER improvement** on weaker models. [PyTorch, LoRA, Transformers, Quantization]

TECHNICAL SKILLS

Systems	Python, C/C++, GCP, Multithreading, REST APIs, Bash, Git, Linux environments, SSH, system-level debugging
MLOps/Deploy	MLflow, TFLite, ONNX Runtime, Docker, Kubernetes, TFExtended, TFServe, CI/CD, HPC, SLURM, Hardware Profiling
AI/ML Core	Machine Learning, Deep Learning, Computer Vision, 3D Perception, Sensor Fusion, ADAS/Autonomous Driving, NLP, Generative AI, Agentic AI, Reinforcement Learning, Self-Supervised Learning, Quantization, Model Optimization (Pruning, QLoRA), Multimodal & Textual RAG Systems, Large-Scale Data Processing & Dataset Curation
Frameworks	PyTorch, TensorFlow, Keras, Hugging Face (Transformers, PEFT), Scikit-learn, OpenCV, NLTK, SciPy, FAISS, OpenSearch, PySpark, NumPy, Pandas, Matplotlib, LangGraph, Quantization-Aware Training, PySpark